

subsequent tick size, so the Commission has evaluated the tradeoffs in response to commenters' concerns.

The Commission acknowledges, as it did in the Proposing Release,¹³⁶⁶ that quoted spreads are not static from day to day. It is possible that a stock could have a narrow quoted spread during an evaluation period, and thus be assigned a \$0.005 tick, and then during the following operative period it could experience points in time where the quoted spread is much wider.¹³⁶⁷ Likewise, a stock could have a wide quoted spread during an evaluation period and therefore be assigned a \$0.01 tick, yet subsequently trade with a narrower spread such that a \$0.005 tick would be beneficial. The extent to which these outcomes occur will depend on the length of the evaluation period, the operative period, and the lag between the two.

The Commission evaluates the tradeoffs mentioned above by computing diagnostic statistics for many different combinations of evaluation and operative periods. The four panels of Table 10 each present different diagnostics: Panel A estimates the fraction of aggregate share volume that is misassigned to a tick of \$0.01, Panel B estimates the fraction of aggregate share volume that is misassigned to a tick of \$0.005, Panel C estimates the rates of false positives, and Panel D estimates the rates of false negatives. Table 10 is similar to Table 10 of the Proposing

¹³⁶⁶ See Proposing Release, supra note 11, at 80324.

¹³⁶⁷ Likewise, two stocks with equal average quoted spreads may not be equally tick-constrained. For example, one stock with a \$0.02 average quoted spread could have a \$0.01 quoted spread 40% of the time while another has a \$0.01 quoted spread 10% of the time. The effect of the Rule on market quality could differ in much the same way as the effects described in this paragraph. Additionally, some commenters inquired about how stock splits and reverse splits would be handled; see SIFMA Letter II at 42 and Virtu Letter II at 18. Stock splits and reverse splits can mechanically affect the bid-ask quoted spread and are not considered in the evaluation periods, thus it is possible that a stock could temporarily be misassigned to a tick size due to a split or reverse split. These effects would be temporary and would rectify in the next evaluation period. See supra section III.C.8 for further discussions of how tick sizes are assigned following a stock split or reverse split.

Release,¹³⁶⁸ which estimated in the context of the proposal the fraction of aggregate volume that would receive a tick reduction yet trade with too many intra-spread ticks during the subsequent three months. Table 10 herein extends this analysis by: computing a wider range of diagnostic statistics, computing the statistics on a rolling basis for a 4.5-year sample, computing the statistics for a wide variety of period lengths, and incorporating a one-month lag between the evaluation period and the implementation of the tick updates as provided for in the adopted amendments.

Each diagnostic in Table 10 is computed for sixteen combinations of evaluation and operative periods—four evaluation period lengths, and four operative period lengths. The evaluation period lengths include: one month, three months (as suggested in the proposal and finalized in the adopted amendments), five months, and twelve months.¹³⁶⁹ The operative period lengths include: one month, three months (as suggested in the proposal), six months (as in the adopted amendments), and twelve months.¹³⁷⁰ The calculations incorporate a one-month lag between the evaluation period and the implementation of the updated tick size – which was not part of the proposal but is part of the adopted amendments. Further, each diagnostic is computed

¹³⁶⁸ See *supra* note 11, at 80324.

¹³⁶⁹ The evaluation periods in this analysis were chosen to cover a range of time periods for both the evaluation period and the operative effective period suggested by commenters who suggested time horizons ranging from monthly updating to annual updating, *see, e.g.*, Pragma Letter at 6 BlackRock Letter at 9, FIA PTG at 2, UBS at 13. Although commenters suggested 6-month evaluation periods, the analysis here considers five-month evaluation period specifically so that the evaluation period would not encompass two tick regimes when combined with a six-month operative period and a one month lagged implementation. For example, consider the six-month operative period spanning May to October (which includes a one month lagged implementation); the next six-month evaluation period would span April to September, thus including two distinct tick regimes. The five-month evaluation period would only span May to September, which is wholly contained in the most recent operative period. An evaluation period that encompasses two tick regimes may be less informative of the appropriateness of the current tick assignment; therefore, Table 10 uses a five-month evaluation period instead of a six-month evaluation period.

¹³⁷⁰ The operative periods were chosen to cover a range of intervals, and to fit evenly into a twelve-month calendar year. In this way, any changes to the tick size would occur in the same month(s) of each year.

for every month from January of 2018 to June of 2022. That is, the Commission simulates the tick assignment procedure on a rolling basis for every month of the 4.5-year sample and computes the diagnostics at every month.¹³⁷¹ Each diagnostic is therefore computed under each of the sixteen combinations of periods and for each of the 54 months in the sample, for a total of 864 calculations per diagnostic. The panels in Table 10 summarize the diagnostics over the 54 months and present a single number for each of the sixteen combinations of periods.

The diagnostics computed in Table 10 are subject to the following caveat: they are estimated from historical data in which the access fee cap was set at 30 mils, and stocks were constrained by the \$0.01 tick. The amendments will reduce the access fee cap to 10 mils for all stocks and will reduce the tick to \$0.005 for stocks that maintain a TWAQS at or below \$0.015 during the evaluation period. These changes are likely to have two opposing effects on quoted spreads: the reduction in the access fee cap will put upward pressure on quoted spreads, while the reduction in the tick will allow quoted spreads to fall below \$0.01 for some NMS stocks.¹³⁷²

This caveat implies that the estimates in Table 10 will be systematically different from the same statistics calculated with realized data—i.e., data after the rule is implemented. However, the purpose of the analysis in Table 10 is to detect patterns in how the diagnostics vary across combinations of evaluation and operative periods. These patterns are likely to be robust to the aforementioned caveat. For example, suppose the reduction in the access fee cap causes a

¹³⁷¹ For example, suppose the month is April 2018. To compute the statistics for an evaluation length of three months and an operative length of six months, the tick size for May 2018 through August 2018 is set by the symbol's TWAQS from January 2018 through March 2018. The statistics in the table are determined by symbols' trading during the May 2018 through August 2018 period (the operative period). This process is repeated for every month from January 2018 to June 2022, and the table summarizes results across all months.

¹³⁷² See supra section VII.D.1.b.ii for a discussion of the effect of the reduction in the tick on quoted spreads, and section VII.D.2 for a discussion of the effect of the reduction in the access fee cap on quoted spreads.

stock's quoted spread to widen by 20 mils; this effect would occur whether the Evaluation Period is three months or twelve months, and whether the operative period is one month or six months, etc. Therefore, Table 10 can still inform the choice of evaluation and operative period. The subsequent discussion of Table 10 will further highlight when a diagnostic may be over- or under-estimated.

Panel A of Table 10 estimates the fraction of aggregate share volume that is misassigned to a tick size of \$0.01. A stock's volume is misassigned to a penny tick if its average TWAQS is above \$0.015 during the most recent evaluation period, yet trades with a TWAQS below \$0.015 in the operative period. This trading volume would have benefited from a lower tick of \$0.005 in those subsequent months and is therefore considered a false negative. The fraction of aggregate share volume that is a false negative is reported in panel A. Further, this fraction is reported for every combination of evaluation period and operative period.

Looking at the false negatives in Panel A, the table demonstrates that shorter periods tend to reduce the percent of volume that is misassigned to a \$0.01 tick. One can pick any evaluation length shown in the table (1, 3, 5, or 12). For that evaluation length, the fraction of aggregate share volume that occurs with a \$0.01 tick and maintains a TWAQS at or below \$0.015 is increasing in the operative period. In other words, more frequent updating provides greater benefits, on average, for any choice of evaluation period. For example, an evaluation period of 3 months with a 1-month operative period will misassign 9.4% of trading volume to a tick of \$0.01; if the same evaluation period is used but the tick is updated annually, then the fraction of misassigned trading increases to 12.9%. Similarly, one can pick any column corresponding to the operative period (1, 3, 6, or 12) and observe that the prevalence of false negatives is increasing

with the length of the evaluation period. These patterns are consistent with commenters' views that more recent (and relevant) data tend to increase the benefits of the amendments.

A tick size can be misassigned in a second way: a stock may be assigned a tick of \$0.005 yet end up trading with an average quoted spread over \$0.015. This trading volume may not fully benefit from the lower tick and is therefore considered a false positive. If the quoted spread widens sufficiently, relative to the quote, then the stock could trade in a range of ticks intra-spread that may harm market quality.¹³⁷³ The fraction of aggregate volume that is a false positive is reported in Panel B of Table 10 for every combination of evaluation period and operative period. This is analogous to Table 10 of the Proposing Release, but with a quoted spread threshold of \$0.015 instead of 10 or 15 ticks to align with the adopted amendments.

Panel B shows that, for the adopted rule choices of a 3-month Evaluation Period and a 6-month operative period, 3.4% of share volume will receive a \$0.005 tick yet trade in an environment with an TWAQS over \$0.015. It is possible that the reduction in the tick size could cause a worse trading environment for some of this fraction of trading volume, compared to what the trading environment could have been had the stock retained a \$0.01 tick. This effect will not be indefinite because, if a stock's quoted spread remains elevated, then at the end of the next evaluation period the stock will be assigned a wider tick—mitigating the negative consequences of having a tick size that is too narrow relative the quoted spread.

¹³⁷³ The empirical analysis in section VII.D.1.b.ii, suggesting that a lower tick size benefits tick-constrained stocks, is an “on average” result. While the Commission expects that a lower tick would on average decrease transaction costs for tick-constrained stocks, the Commission cannot rule out the possibility that for some of these stocks, a smaller tick could lead to wider quoted spreads. For these stocks, if quoted spreads increase to a sufficient degree, then the stock could be re-assigned a wider tick after the next evaluation period.

The false positive statistics in Panel B of Table 10 further show that less frequent tick updating tends to result in more volume trading with a smaller tick yet a relatively wide quoted spread. This can be seen by the increased prevalence of false positives as one moves left-to-right along a row—for any chosen evaluation period, a longer operative period results in more volume trading at a low-tick yet relatively wide quoted spread. This pattern is consistent with commenters' views that more frequent updating is better at adapting to changing market trends and stock behavior. Similarly, moving down any given column shows that increasing the evaluation period tends to reduce the amount of volume in low-tick stocks with wide quoted spreads; this pattern is consistent with commenters' views that short evaluation periods may, in some circumstances, assign stocks to the \$0.005 tick on the basis of transient events, and that these stocks may not be able to sustain the tick reduction.

The estimates of false positives in Panel B of Table 10 are likely to be over-estimates for two reasons. First, the threshold for tick misassignment is chosen at a quoted spread of \$0.015, which corresponds to three intra-spread ticks. The above analysis on the Tick Size Pilot indicates that the trading environment does not deteriorate until the number of intra-spread ticks is well above three. Hence, the chosen threshold is conservative. Second, and as previously discussed, the estimates in Panel B are constructed from historical data in which stocks were constrained by the \$0.01 tick. Once these amendments are implemented the stocks assigned a \$0.005 tick will be able to trade at quoted spreads below \$0.01; this will have a mechanical effect of lowering the stocks' TWAQS and thereby keeping more volume under the \$0.015 quoted spread threshold.

By comparing the magnitudes of the false negatives and false positives in Panels A and B of Table 10, we can assess the relative importance of the two ways in which a tick may be misassigned. The false negatives are generally substantially larger and exhibit greater variation

within the table.¹³⁷⁴ This implies that the incremental effect of the choice of period length will be greater for the share of volume that is misassigned to a \$0.01 tick than for volume that is misassigned to a \$0.005 tick. For example, the difference between the highest and lowest fraction of false negatives is 12.3%—this is the increase in aggregate share volume at a misassigned \$0.01 tick when moving from 1-month periods to 12-month periods—while the difference between the highest and lowest fraction of false positives is only 1.8% of aggregate share volume.¹³⁷⁵

An advantage of Panels A and B of Table 10 is that they estimate the prevalence of false negatives and positives as fractions of aggregate share volume; these panels therefore show the amount of aggregate trading that is misassigned to a tick. A disadvantage of Panels A and B is that they do not condition on the aggregate level of quoted spreads. Some combinations of evaluation and operative periods may do relatively well when aggregate quoted spreads are high, and some combinations may do better when aggregate quoted spreads are low.¹³⁷⁶

¹³⁷⁴ If false positives are over-estimated, as discussed in the previous paragraph, then the relative importance of false negatives becomes greater still.

¹³⁷⁵ The greater prevalence of false negatives may be due to the existing \$0.01 tick size effectively censoring the observed quoted spreads at the penny tick. For symbols near the \$0.015 threshold, the prevalence of false positives and negatives should be approximately equal—*i.e.*, a symbol with a quoted spread of \$0.0149 is as likely to cross over the \$0.015 threshold (and be a false positive) as a symbol with a quoted spread of \$0.0151 is likely to cross under the \$0.015 threshold (and be a false negative). When we move away from the threshold, however, the low-tick symbols are more likely to be constrained by the penny tick—if we see a TWAQS of \$0.012, the true market-clearing quoted spread —*i.e.*, unconstrained by the minimum tick—is likely lower than \$0.012 because the TWAQS is censored at \$0.01; if we see a TWAQS of \$0.018, however, this censoring is less important. Therefore, the likelihood that a symbol with a TWAQS of \$0.012 crosses over the \$0.015 threshold (and is recorded as a false positive) is lower than the likelihood that a symbol with a TWAQS of \$0.018 crosses under the \$0.015 threshold (and is recorded as a false negative).

¹³⁷⁶ To take an extreme example, suppose every stock's quoted spread increases by \$0.10 immediately after new ticks are assigned. This implies that no stock should have a tick of \$0.005. In this case, the evaluation and operative periods that assign the least amount of trading to the \$0.005 would do best. Conversely, suppose that quoted spreads fall by half immediately after new ticks are assigned; in this case, the periods that assign more trading to the \$0.005 tick would generally do better.

Panel C and D of Table 10 address the aforementioned disadvantage of Panels A and B by estimating rates of false negatives and positives. The false negative rate conditions on the amount of low-quoted spread volume, while the false positive rate conditions on the amount of high-quoted spread volume.¹³⁷⁷ Specifically, the rate of false negatives is measured as the amount of share volume that occurs with a tick of \$0.01 and a TWAQS under \$0.015, divided by the total amount of share volume that occurs with a TWAQS under \$0.015. Analogously, the rate of false positives is measured as the amount of share volume that occurs with a tick of \$0.005 and a TWAQS over \$0.015, divided by the total amount of share volume that occurs with a TWAQS over \$0.015. In the context of the evaluation period, a low false negative rate signifies effective identification of stocks that would benefit from a tick reduction, while a low false positive rate suggests effective avoidance of assigning a reduced tick to stocks that would not benefit from it. In contrast to Panels A and B, patterns in Panels C and D of Table 10 are likely to be more robust to market-wide changes in quoted spreads.

Panel C of Table 10 presents results for the false negative rates across 16 combinations of evaluation periods and operative periods. The pattern for false negative rates is similar to the fraction of share volume inappropriately assigned a \$0.01 tick in Panel A—as the period lengths shorten, the false negative rates decrease. For example, an evaluation period of 3 months with monthly updating typically fails to assign a tick reduction to 12.8% of volume that would benefit from it. If the same evaluation period is used but the tick is updated annually, then the rule would fail to assign a tick reduction to 19.3% of volume that would benefit from it. The pattern holds moving down columns and moving across rows from shorter to longer periods. This lends

¹³⁷⁷ In the statistics and medicine literature, the false positive rate is related to a test's specificity, while a false negative rate is related to a test's sensitivity.

support to commenters' views that more recent data—from shorter evaluation periods and more frequent updating—tends to do a better job at assigning a low tick to stocks that will benefit from it in the operative period.

False positive rates are reported in Panel D. The pattern for false positive rates is consistent with results in Panel B on the fraction of volume inappropriately assigned a \$0.005 tick—shorter evaluation periods and longer operative periods tend to have higher false positive rates, indicating that more trading is potentially harmed from having too narrow of a tick. For every column, the highest false positive rate occurs with an evaluation length of 1 month, and the lowest false positive rate occurs with an evaluation length of 12 months. This indicates that short evaluation periods may put more weight on transient events, as some commenters stated. Similarly, for every row the false positive rate is highest with an operative period of 12 months, and the rate is lowest with an operative period of 1 month. This indicates that infrequent updating increases the risk that a stock is stuck at an inappropriately low tick for many months.

The contrasting patterns in the rates of false positives and negatives illustrates a tradeoff highlighted by commenters (and discussed at the beginning of this section). A short evaluation period uses the most recent and relevant information, which on average reduces the false negative rate; however, a short evaluation period also places more emphasis on short-term and volatile events, which raises the false positive rate.

Table 10: Effect of the evaluation period on inappropriate tick assignment					
Panel A: Fraction of aggregate share volume assigned a \$0.01 tick with a subsequent TWAQS below \$0.015 (<u>i.e.</u> , false negatives)					
		Length of operative period in months			
		1	3	6	12
Length of evaluation	1	8.5%	9.4%	10.7%	12.9%
period in months	3	11.4%	12.4%	13.7%	15.8%

	5	13.1%	14.1%	15.3%	17.3%
	12	17.0%	17.8%	18.9%	20.8%
Panel B: Fraction of aggregate share volume assigned a \$0.005 tick with a subsequent TWAQS above \$0.015 (<u>i.e.</u> , false positives)					
Length of operative period in months					
		1	3	6	12
Length of evaluation period in months	1	3.1%	3.3%	3.8%	4.3%
	3	2.6%	2.9%	3.4%	3.8%
	5	2.6%	2.9%	3.2%	3.6%
	12	2.5%	2.7%	2.9%	3.2%
Panel C: False negative rates					
Length of operative period in months					
		1	3	6	12
Length of evaluation period in months	1	12.8%	14.2%	16.2%	19.3%
	3	17.4%	18.8%	20.6%	23.6%
	5	19.9%	21.2%	23.0%	25.9%
	12	25.7%	26.9%	28.6%	31.2%
Panel D: False positive rates					
Length of operative period in months					
		1	3	6	12
Length of evaluation period in months	1	8.5%	9.4%	10.9%	12.6%
	3	7.2%	8.2%	9.6%	11.2%
	5	7.1%	8.1%	9.3%	10.6%
	12	6.8%	7.5%	8.4%	9.5%
^a For every month from January 2018 to June 2022, the Commission simulates the tick assignment procedure under 16 combinations of evaluation and operative period lengths. The evaluation period determines the number of prior months to use when averaging each stock's quoted spread; a TWAQS of \$0.015 or below during the evaluation period causes the stock to receive a tick of \$0.005 during the subsequent tick assignment interval. The operative period determines the length of each tick assignment. All the statistics in the tables are computed using data beginning one month after the evaluation period. For example, suppose the month is April 2018. To compute the statistics for an evaluation					